

Informe de experimento: results_validation_gpt-4o-mini_stress

Generado el 2026-09-16 11:20 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-4o-mini_stress (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_gpt-4o-mini_stress
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-4o-mini
- Métodos: pfagent
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=['stress'] (1 report)
- Items por dificultad: stress=40
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 10:44
- Costo total: 0.1325 USD (40 items; costo medio 0.00331 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=836246
- Solved global: 60.0% (24/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion`: **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
pfagent	case1440	40	100.0 (40/40)	–	0.00	60.0 (24/40)	100.0 (40/40)	0	100.0	100.0 (24/24)	4.5	209060.0	0.0 (0/24)	100.0 (16/16)	–	0.1325

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	stress
pfagent	100.0 (40/40)

Solved por dificultad (%, items resueltos / items)

Method	stress
pfagent	60.0 (24/40)

Verificación final (pfagent: $V(x,c,z,y)$ aplicado a la respuesta final)

`pass_first` = pasó al primer intento; `pass_retry` = pasó tras el único reintento con el veredicto añadido; `abstained` = falló dos veces, respuesta forzada a fallo declarado sin números; `retry_mutation` = el reintento intentó una herramienta que muta la red (bloqueada, nunca ejecutada; cuenta como abstención) —ver “Integridad del reintento” abajo. N = items con `final_gate` activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained	retry_mutation
pfagent	40	0.0 (0/40)	0.0 (0/40)	100.0 (40/40)	0.0 (0/40)

Integridad del reintento

De 40 ítem(s) que llegaron a un reintento, 0 intentaron una herramienta que muta la red (bloqueada) en vez de corregir el reporte; se cuentan como abstención, no como “pasó”. El mensaje de reintento ya no sugiere la corrección específica (p. ej. “reconnect...”) y bloquea estructuralmente `modify_load`, `disconnect_line`, `reconnect_line`, `apply_remedial_action` y `load_case` mientras el reintento está en efecto.

Tipos de error de formulación (conteo de items)

Method	ok
pfagent	40

Errores de ejecución por método (conteo de items con error)

(ningún item con *error*)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (`claimed_success_on_failure`, `stale_state`, `abstained`). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-stress-003-s0, case14, difficulty=stress, seed=1911784258): Load the 14-bus test case (case14), disconnect the line between
↳ bus 1 and bus 2, increase the load at bus 3 to 282.6 MW and 57 Mvar (three times its nominal value), and finally rerun the power flow,
↳ then report the bus with the lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
↳ 282.6, "q_mvar": 57.0}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
↳ 282.6, "q_mvar": 57}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 1, "to_bus": 2})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -34.6873802729158, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true}], "line_flows": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false, "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true, "violation_type": null}], "thermal_violations": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 3 ---
LLM: The power flow analysis did not converge after the modifications made to the system.

1. **Summary**: The power flow failed to converge after disconnecting the line between bus 1 and bus 2 and increasing the load at bus 3.
2. **Key numbers**: No numerical values can be reported due to non-convergency ...[225 chars más]
--- Ronda 3 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": false, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true}], "line_flows": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false, "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true, "violation_type": null}], "thermal_violations": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 4 ---
LLM: The power flow analysis still did not converge after the modifications made to the system.

1. **Summary**: The power flow failed to converge after disconnecting the line between bus 1 and bus 2 and increasing the load at bus 3.
2. **Key numbers**: No numerical values can be reported due to non-conv ...[242 chars más]
--- Respuesta final ---
No numerical result is reported. Verification failed: the last power flow did not converge after setting the load at bus 3 to 282.6 MW.
↳ This is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=--; faithful=100.0 %; solved=True (declared_failure); flags=-; error=-; tokens=13620;
↳ cost_usd=0.0022; wall=5.8 s
```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
↳ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true}], "line_flows": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false, "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true, "violation_type": null}], "thermal_violations": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=9.78e-09 MW)
--- Ronda 2 ---
LLM: The IEEE 14-bus system has been loaded, and the transformer branch between bus 7 and bus 8 has been disconnected.

- **Total Load:** 252.66 MW
- **Total Generation:** 265.46 MW
- **Total Losses:** 12.81 MW
- **Voltage Range:** 1.01 to 1.07 p.u.

**Violations:**
- Overvoltage at buses 1, 6, 11, and 1 ...[145 chars más]
--- Ronda 2 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true}], "line_flows": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false, "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -56.24, "is_violation": true, "violation_type": null}], "thermal_violations": [{"line_id": 1, "p_mw": 0.0, "q_mvar": 0.0, "p_loss_mw": 0.0, "q_loss_mvar": 0.0, "p_flow_mw": 0.0, "q_flow_mvar": 0.0, "p_loss_pu": 0.0, "q_loss_pu": 0.0, "p_flow_pu": 0.0, "q_flow_pu": 0.0}], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=9.78e-09 MW)
--- Ronda 3 ---
LLM: The IEEE 14-bus system has been loaded, and the transformer branch between bus 7 and bus 8 has been disconnected. However, this action
↳ has isolated bus 8, resulting in it having no valid voltage.
```

```

- **Total Load:** 252.66 MW
- **Total Generation:** 265.46 MW
- **Total Losses:** 12.81 MW
- **Voltage ...[360 chars más]
--- Respuesta final ---
No numerical result is reported. Verification failed: bus 8 has no valid voltage because disconnecting the branch between bus 7 and bus 8
↳ isolated it from the slack bus, so the network is split. Suggested next step: reconnect the branch between bus 7 and bus 8, or analyse
↳ the connected part separately. This is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=False (failure_not_declared);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=14022; cost_usd=0.0023; wall=6.0 s

```

Ejemplo: no resuelto

```

request (case14-stress-001-s0, case14, difficulty=stress, seed=827308000): Load the 14-bus test case (case14), disconnect the line between
↳ bus 2 and bus 4, disconnect the transformer branch between bus 7 and bus 8, and finally rerun the power flow, then list every line
↳ whose loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 2, "to_bus": 4}) -> disconnect_line({"from_bus": 7,
↳ "to_bus": 8}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 2, "to_bus": 4}) -> disconnect_line({"from_bus": 7,
↳ "to_bus": 8}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 270.1361769951196,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 2, "to_bus": 4})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.717168458565059, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5946 chars más]
gate: PASS (reasons=-, mismatch=3.08e-08 MW)
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.733791919883227, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5459 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=8.78e-08 MW)
--- Ronda 2 ---
LLM: The power flow analysis has been completed after disconnecting the specified lines.

1. Summary: The power flow converged successfully after the modifications.
2. Key numbers:
- Total Load: 270.14 MW
- Total Generation: 287.42 MW
- Total Losses: 17.28 MW
- Voltage Range: 0.9966 to 1.07 ...[222 chars más]
--- Ronda 2 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.733791919883227, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5459 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=8.78e-08 MW)
--- Ronda 3 ---
LLM: The power flow analysis has been re-evaluated after the network modifications.

1. Summary: The power flow converged successfully, but bus 8 remains isolated with no valid voltage due to the disconnection of the line
↳ between bus 7 and bus 8.
2. Key numbers:
- Total Load: 270.14 MW
- Total Gen ...[391 chars más]
--- Respuesta final ---
No numerical result is reported. Verification failed: bus 8 has no valid voltage because disconnecting the branch between bus 7 and bus 8
↳ isolated it from the slack bus, so the network is split. Suggested next step: reconnect the branch between bus 7 and bus 8, or analyse
↳ the connected part separately. This is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=False (failure_not_declared);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=21637; cost_usd=0.0034; wall=7.3 s

```

```

1. Summary: The power flow converged successfully, but bus 8 remains isolated with no valid voltage due to the disconnection of the line
↳ between bus 7 and bus 8.
2. Key numbers:
- Total Load: 270.14 MW
- Total Gen ...[391 chars más]
--- Respuesta final ---
No numerical result is reported. Verification failed: bus 8 has no valid voltage because disconnecting the branch between bus 7 and bus 8
↳ isolated it from the slack bus, so the network is split. Suggested next step: reconnect the branch between bus 7 and bus 8, or analyse
↳ the connected part separately. This is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=False (failure_not_declared);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=21637; cost_usd=0.0034; wall=7.3 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- El gate verificó 142 resultados de flujo, 81 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de **pfagent**: 60.0% (24/40).
- Formulación exacta agregada sobre métodos con tools: 100.0% (40/40).
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- De 24 items en estado de fallo, 24 lo declararon de forma segura (SFR 100.0% (24/24)).
- 0 de 40 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.) media con tools: 100.0%.

- 40 ítems esperaban un resultado no convergido (`non_converged=24`, `islanded=16`); el método lo declaró correctamente (`solved`) en 60.0 % (24/40), con `solved_reason declared_failure=24`.
- Costo total 0.1325 USD; el método más caro fue `pfagent` con 0.1325 USD (100 %).
- Métodos con tools: 4.55 tool calls medias por ítem (máximo 7).

6. Archivos

Rutas absolutas bajo `results_validation_gpt-4o-mini_stress.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/report.json    [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_stress/traces
```